# Exercise Sheet 9

## Diffusion and Epidemics

5942UE Network Science

26 March 2026

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# 9.A Diffusion as a Network Process

# Exercise 9.A.1: Simple Diffusion Trace

Directed graph: $A \to B, A \to C, B \to D, C \to D, D \to E$.

Node $A$ is active at $t = 0$. At each step, an active node activates all out-neighbours.

1. Trace the active set at $t = 1, 2, 3$.
2. Which node is never reached? Why?
3. If you add the edge $B \to C$, does anything change for this seed?

# Exercise 9.A.2: Structural Bottlenecks

Two 8-node graphs with seed $S$:

- **Star:** $S$ connected to every other node.
- **Line:** $S - A - B - C - D - E - F - G$.

1. How many steps until all nodes are active in each?
2. Where is the structural bottleneck?
3. A single edge fails. Which topology is more affected?

# Exercise 9.A.3: Centrality and Diffusion Speed

Load the karate club graph and BFS-diffuse from the highest-degree node.

1. How many steps until all 34 nodes are active?
2. Repeat from the lowest-degree node. How does the curve change?
3. What does this say about hubs?

# Exercise 9.A.4: Paths and Reachability

Company-wide email cascade from CEO ($C$) must reach 500 employees. The network has a core-periphery structure: 50 managers in a dense core, each linked to $\approx 10$ peripheral employees.

1. Does a message from $C$ to a peripheral employee require passing through the core?
2. If 5 of 50 managers are absent, what fraction of periphery is potentially unreachable?
3. How could the company become more resilient?

# 9.B Epidemic Models and Simple Contagion

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# Exercise 9.B.1: SIR Trace

SIR rules (synchronous): $S \to I$ if any infected neighbour, with probability $\beta$. $I \to R$ at the next step (single-step infectious period).

Path graph $A - B - C - D - E$ with $B$ infected at $t = 0, \beta = 1.0$.

1. Trace each node's state at $t = 0, 1, 2, 3, 4$.
2. When does $E$ become infected?
3. With $\beta = 0.5$, what is $P(\text{C not infected at } t = 1)$?

# Exercise 9.B.2: Structural Effect of Topology

SIR with $\beta = 1.0$, one-step recovery, seed = node 1.

- **Line L:** $1 - 2 - 3 - 4 - 5 - 6$
- **Star S:** node 1 is the hub of $2, 3, 4, 5, 6$

1. Maximum simultaneously infected in L?
2. Maximum simultaneously infected in S?
3. Which produces a sharper epidemic peak? Implications?

# Exercise 9.B.3: SIR Simulation in NetworkX

Implement synchronous SIR on the karate club graph and analyse it.

1. At what step does the epidemic peak?
2. What fraction ends in $R$ (ever infected)?
3. Vary $\beta$ from 0.1 to 0.8. Where does the epidemic reliably infect $> 80$?

# Exercise 9.B.4: Simple vs. Complex Contagion

For each scenario, decide whether it is **simple** (one contact sufficient) or **complex** (multiple confirmations needed):

(a) A cold virus. (b) Adopting a new software tool only after three colleagues recommend it. (c) Sharing a news article posted by one friend. (d) Joining a protest after pressure from multiple community members. (e) Updating a belief after seeing a claim from several independent sources.

For (b) and (d), how must network structure differ from simple contagion?

# Wrap-Up

- diffusion is bounded by reachable paths; redundant paths don't help unless primary paths fail
- bottlenecks and hubs control epidemic dynamics
- the same $\beta$ produces dramatically different curves on different topologies
- complex contagion reverses the role of bridges and clusters compared to simple contagion